FF-B-584E
AMENDMENT-1
October 19, 1977

FEDERAL SPECIFICATION

BOLTS, FINNED NECK; KEY HEAD; MACHINE:
RIBBED NECK; SQUARE NECK: TEE HEAD

This amendment which forms a part of Federal Specification FF-B-584E, dated June 17, 1976, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

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Delete Table I in its entirety and substitute the following:

FSC 5306

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TABLE I. Mechanical and chemical properties of carbon and alloy steel bolts

Grade designation	Nominal size diameter inches	Full size bolts, screws, studs, sems: Proof load (Stress) psi	Full size bolts, screws, studs, sems: Tensile strength (Stress) min psi	Machine test specimens of bolts, screws, and studs: Yield strength (Stress) min psi	Machine test specimens of bolts, screws, and studs: Tensile strength (Stress) min psi	Machine test specimens of bolts, screws, and studs: Elongation min, percent	Machine test specimens of bolts, screws, and studs: Reduction of area, min, percent	Surface hardness: Rockwell 30N max.	Core hardness: Rockwell Min	Core hardness: Rockwell Max.	Grade identification marking
1	1/4 through 1-1/2	33,000	60,000	36,000	60,000	18	35		B70	B100	None
2	1/4 through 3/4	55,000	74,000	57,000	74,000	18	35		B80	B100	
	Over 3/4 to 1-1/2	33,000	60,000	36,000	60,000	18	35		B70	B100	None
4	1/4 through 1-1/2	-	115,000	100,000	115,000	10	35		C22	C32	None
5	1/4 through 1	85,000	120,000	92,000	120,000	14	35	54	C25	C34	
	Over 1 to 1-1/2	74,000	105,000	81,000	105,000	14	35	50	C19	C30	
5.1	No. 6 through 3/8 Sems only, through 5/8	85,000	120,000	-	-	-	-	59.59	C25	C40	
5.2	1/4 through 1	85,000	120,000	92,000	120,000	14	35	56	C26	C36	
7	1/4 through 1-1/2	105,000	133,000	115,000	133,000	12	35	54	C28	C34	
8	1/4 through 1-1/2	120,000	150,000	130,000	150,000	12	35	58.6	C33	C39	
8.1	1/4 through 1-1/2	120,000	150,000	130,000	150,000	10	35		C32	C38	None
8.2	1/4 through 1	120,000	150,000	130,000	150,000	10	35	61	C35	C42	

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Chemical composition

Grade	Material and treatment	Element, percent					
		C		Mn	P	S	B
		Min	Max.	Min	Max.	Max.	Min
1	Low or medium carbon steel	--	0.55	--	0.048	0.058	--
2	Low or medium carbon steel	--	0.28	--	0.048	0.058	--
4	Medium carbon cold drawn steel	--	0.55	--	0.048	0.058	--
5	Medium carbon steel, quenched and tempered	0.28	0.55	--	0.048	0.058	--
5.1	Low or medium carbon steel, quenched and tempered	0.15	0.30	--	0.048	0.058	--
5.2	Low carbon martensite steel, fully killed, fine grain, quenched and tempered	0.15	0.25	0.74	0.048	0.058	0.0005
7	Medium carbon alloy steel, quenched and tempered	0.28	0.55	--	0.040	0.045	--
8	Medium carbon alloy steel, quenched and tempered	0.28	0.55	--	0.040	0.045	--
8.1	Elevated temperature drawn steel - medium carbon alloy or SAE 1541 (formerly SAE 1041)	0.28	0.55	--	0.048	0.058	--
8.2	Low carbon martensite steel, fully killed, fine grain, quenched and tempered	0.15	0.25	0.74	0.048	0.058	0.0005

MILITARY INTEREST:

Custodians:

Army - WC
Air Force - 99

Review activities:

Army - MI, MU
Navy - SH

User activities:

Army - AT, AV
Navy - MC

Civil Agency Coordinating Activities:

AGR - AFS
GSA - FSS
JUSTICE - FPI

Preparing activity:

Navy - YD

Project No. 5306-0590

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